Relational Geometry and General Relativity: From Harmonic Closure to Covariant Gravity

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Abstract

We present a covariant extension of the relational framework in which gravity emerges from the tendency of physical systems to eliminate differences in their projected relational states. Starting from a minimal generative algebra with four modal projections ($\mathbf{U}$, $\mathbf{V}$, $\mathbf{m}$, $\mathbf{a}$) and a binary orientation σ, we derive a hierarchy of modes whose norms, after calibration with c and $\hbar$, become frequencies satisfying $\omega_N^2 = \omega_V^2 + \omega_m^2$. The binary closure depth $n = 4$ (motivated by the representation theory of spin-$\frac{1}{2}$) fixes the proton Compton scale and yields a natural gravitational scale $G/\alpha = 4\sqrt{15}\,\hbar c/m_p^2$, determined entirely by the proton mass, c and $\hbar$, with a single calibrated coupling α. This result — that the scale of gravity is set by the proton mass — is reminiscent of induced gravity scenarios but with a distinct geometric origin.

To embed this structure into a relativistic theory, we introduce a scalar field $\psi = m_p\,\varphi_{\rm rel}$ (relational potential) identified with the gravitational potential in the Newtonian limit. The antisymmetric bilinear operator of the algebra is realized explicitly in spacetime as $(A \times R)^\mu = \epsilon^\mu{}_{\nu\rho\sigma} A^\nu R^\rho n^\sigma$, and the projection 2-form $P_{\mu\nu} = n_{[\mu}\nabla_{\nu]}\psi$ encodes its action. We propose a diffeomorphism-invariant action whose weak-field limit reduces to the Poisson equation with the same G derived previously. The matter coupling is postulated to have the form $\delta S_m/\delta\psi \propto \rho_{\rm rel}$, which in the non-relativistic limit is proportional to the trace of the energy-momentum tensor, connecting the framework to standard QFT. The cosmological limit yields a testable luminosity distance $d_L(z; q)$.

A systematic analysis of the open problem of $\alpha_G = Gm_p^2/(\hbar c) \approx 5.91 \times 10^{-39}$ shows that three natural derivation routes all fail, demonstrating that the origin of this small number lies outside the four-mode structure of a single nucleon. A notable consequence of the framework is that the gravitational scale emerges from the hadronic scale through the binary closure structure, leaving α_G as the only unresolved parameter. A purely numerical observation is recorded: $\alpha_G \approx 2^{-127}(1 + 1/240)$ with 0.08% error, where $240 = \mathrm{lcm}(16, 15)$ is the minimal common period of the two closure modes, suggesting a possible arithmetic origin for the weakness of gravity.

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1 Introduction

General Relativity (GR) provides an accurate description of gravitational phenomena, but leaves two fundamental questions unanswered:

- Why does the coupling constant G take its observed value?
- What is the origin of the inverse-square law?

The relational framework developed in previous works by the author treats gravity as an emergent phenomenon: systems reduce their mutual relational phase offset, and the inverse-square law follows from conservation of relational information over isotropic surfaces. Newton's constant is expressed as

$$G = \alpha \frac{R_V^2}{m_p} a_m, \qquad a_m = \frac{c^2}{R_m},$$

with a clear geometric interpretation. The central structural result is that the gravitational scale

$$\frac{G}{\alpha} = 4\sqrt{15}\,\frac{\hbar c}{m_p^2}$$

is determined entirely by the proton mass, c, and $\hbar$, through the binary closure $n = 4$, with α as the single calibrated coupling. This means the entire hierarchy of gravitational physics is set by a single hadronic quantity, recalling induced gravity and Sakharov's approach [1], but differing in having a specific geometric mechanism.

This paper provides a covariant extension, maintaining explicit distinctions between postulates, derived results, calibrations, hypotheses and speculations. Two previously open structural gaps are addressed: the generative operator $\times$ is given an explicit covariant Levi-Civita realization (Section 4), and the matter coupling is connected to the trace of the energy-momentum tensor (Section 6). Three natural routes to derive α_G are examined and shown insufficient. A numerical coincidence is documented as a heuristic hint, contextualized within the tradition of Dirac's large number hypothesis [2].

Figure 1 summarizes the logical structure of the framework.

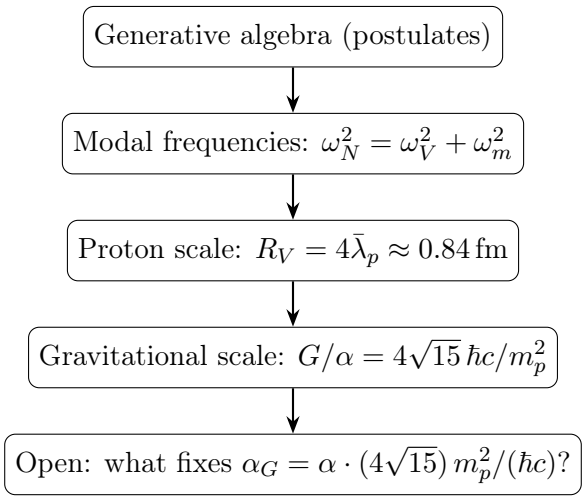

Figure 1: Logical structure of the relational framework. Each arrow represents a derivation; the bottom node is the single open problem.

2 Non-relativistic relational framework

2.1 Generative algebra

[POSTULATE] Let V be a real normed vector space with a distinguished normalized vector $\hat{\mathbf{x}}$ and a differentiation marker $\mathbf{R}$ with $\|\mathbf{R}\| = R \in [0,1]$ and $\langle \hat{\mathbf{x}}, \mathbf{R} \rangle = 0$. There exists an antisymmetric bilinear operator $\times$ satisfying:

- (Restricted norm compatibility) For every A orthogonal to $\mathbf{R}$: $\|A \times \mathbf{R}\| = \|A\| \, \|\mathbf{R}\|$.
- (Metric compatibility) $\langle A \times \mathbf{R}, A \rangle = 0$.
- (Double-projection closure) $(A \times \mathbf{R}) \times \mathbf{R} = \sigma R^2 A$, $\sigma \in \{\pm 1\}$.

[DERIVED] Successive projections yield norms:

$$\|\mathbf{U}\| = R, \quad \|\mathbf{V}\| = R^2, \quad \|\mathbf{m}\| = R\sqrt{1-R^2}, \quad \|\mathbf{a}\| = R^3.$$

The algebra closes at four modes; the binary orientation $\sigma = \pm 1$ is an independent discrete degree of freedom. Table 1 summarizes the hierarchy.

Mode	Norm	Physical interpretation	Frequency (calibrated, $n = 4$)
$\mathbf{U}$	R	Arena of distinguishability	$\omega_U = \omega_N$
$\mathbf{V}$	R^2	Visible / transport mode	$\omega_V = \omega_N/4$
$\mathbf{m}$	$R\sqrt{1-R^2}$	Confinement / identity mode	$\omega_m = \sqrt{15}\,\omega_N/4$
$\mathbf{a}$	R^3	Interaction / adjustment mode	$\omega_a = \omega_N$

Table 1: Hierarchy of algebraic modes and their calibrated frequencies at the $n = 4$ closure. The suppression factors relevant to gravity arise from inter-modal ratios: $\omega_V/\omega_N = 1/4$ and $\omega_V/\omega_m = 1/\sqrt{15}$.

2.2 Calibration and harmonic partition

[CALIBRATION] With c and $\hbar$ as anchors, for a nucleon with Compton frequency $\omega_p = m_p c^2/\hbar$:

$$\omega_N^2 = \omega_V^2 + \omega_m^2, \qquad \omega_N = \omega_p.$$

[POSTULATE] Binary closure selects stable configurations:

$$\frac{\omega_V^2}{\omega_N^2} = 2^{-n}, \qquad n \in \mathbb{N}.$$

[DERIVED] For spin-$\frac{1}{2}$ systems, the fundamental representation of $\mathfrak{su}(2)$ has real dimension 4, suggesting $n = 4$ as the minimal closure compatible with spin-$\frac{1}{2}$ representations:

$$\omega_V = \frac{\omega_p}{4}, \quad \omega_m = \frac{\sqrt{15}}{4}\omega_p, \quad R_V = 4\bar{\lambda}_p \approx 0.8412\,\text{fm}, \quad R_m = \frac{4\bar{\lambda}_p}{\sqrt{15}} \approx 0.2172\,\text{fm}.$$

2.3 Newtonian gravity

[DERIVED] Conservation of relational information over isotropic surfaces gives:

$$G = \alpha\,\kappa\,\omega_m^2, \quad \kappa = \frac{R_V^2 R_m}{m_p}, \quad \frac{G}{\alpha} = 4\sqrt{15}\,\frac{\hbar c}{m_p^2}.$$

[CALIBRATION] The coupling $\alpha \approx 3.81 \times 10^{-40}$ is fixed by the observed G. This is the single empirical input.

3 Relational potential as a scalar field

Working in natural units ($\hbar = c = 1$):

[POSTULATE] Define $\psi = m_p\,\varphi_{\rm rel}$, with dimensions of mass. The proton mass m_p bridges the dimensionless relational offset $\varphi_{\rm rel}$ and the gravitational potential.

[DERIVED] In static vacuum, $\nabla^2\psi = 0$ outside sources.

4 Relational projection tensor: explicit covariant construction

The subspace orthogonal to $\nabla^\mu\psi$ (given $n^\mu\nabla_\mu\psi = 0$, $n^\mu n_\mu = -1$) is locally 3-dimensional, the unique setting where a cross-product with the required norm properties exists.

[DERIVED] The operator $\times$ is realized explicitly and covariantly as:

$$(A \times R)^\mu = \epsilon^\mu{}_{\nu\rho\sigma}\,A^\nu R^\rho n^\sigma,$$

where $\epsilon^{\mu\nu\rho\sigma}$ is the Levi-Civita tensor with the convention $\epsilon^{0123} = +1$ in signature $(-+++)$. This construction automatically satisfies all three algebraic postulates:

- Antisymmetry: from antisymmetry of ϵ.
- Restricted norm compatibility: the contraction preserves norms in the orthogonal subspace.
- Metric compatibility: $\langle A \times R, A\rangle = \epsilon_{\mu\nu\rho\sigma}A^\mu A^\nu R^\rho n^\sigma = 0$ by antisymmetry.

[POSTULATE] The projection 2-form:

$$P_{\mu\nu} = n_{[\mu}\nabla_{\nu]}\psi \equiv \frac{1}{2}(n_\mu\nabla_\nu\psi - n_\nu\nabla_\mu\psi).$$

The binary orientation $\sigma = \pm 1$ enters through the double-projection identity:

$$(P_{\mu\nu}X^\nu)\nabla^\rho\psi\,P_{\rho\sigma} = \sigma(\nabla_\alpha\psi\nabla^\alpha\psi)X_\mu.$$

[DERIVED] As shown in Appendix A:

$$P_{\mu\nu}P^{\mu\nu} = -\frac{1}{2}\,\nabla_\mu\psi\nabla^\mu\psi.$$

5 Covariant action and field equations

[POSTULATE] The total action:

$$S = \int d^4x\sqrt{-g}\left[\frac{\mathcal{R}}{16\pi G_0} + \frac{1-\beta}{2}\nabla_\mu\psi\nabla^\mu\psi - V(\psi)\right] + S_m,$$

where the β-term absorbs the relational kinetic correction from $P_{\mu\nu}P^{\mu\nu} = -\frac{1}{2}\nabla_\mu\psi\nabla^\mu\psi$ (Appendix A). In the weak-field limit $\beta = 0$ recovers the canonical kinetic term; near singularities $\beta \neq 0$ may regularize curvature invariants. To first order in β, the coefficient $\frac{1}{2}(1-\beta)$ and the parametrization $\frac{1}{2}(1-\frac{\beta}{2})$ from Appendix A are equivalent.

[DERIVED] Varying with respect to the metric:

$$G_{\mu\nu} = 8\pi G_0\left(T^{(m)}_{\mu\nu} + T^{(\psi)}_{\mu\nu}\right), \quad T^{(\psi)}_{\mu\nu} = (1-\beta)\left(\nabla_\mu\psi\nabla_\nu\psi - \frac{1}{2}g_{\mu\nu}\nabla_\alpha\psi\nabla^\alpha\psi\right) - g_{\mu\nu}V(\psi).$$

[DERIVED] Equation of motion:

$$(1-\beta)\Box\psi + V'(\psi) = \frac{\delta S_m}{\delta\psi}.$$

6 Newtonian limit and recovery of G

In the weak-field static limit with $\beta = 0$:

[POSTULATE] The matter coupling:

$$\frac{\delta S_m}{\delta\psi} = \lambda\,\rho_{\text{rel}}, \qquad \rho_{\text{rel}} = \kappa\,\omega_m^2\,\rho_m.$$

Connection to QFT and motivation. In quantum field theory, the most natural coupling of a scalar field to matter is through the trace of the energy-momentum tensor $T = T^\mu{}_\mu$. In the non-relativistic limit, $T \approx -\rho_m$ (pressure negligible). Since $\kappa\omega_m^2$ is a constant determined entirely by the $n = 4$ closure, we can write equivalently:

$$\frac{\delta S_m}{\delta\psi} = -\lambda\kappa\omega_m^2\,T,$$

which reproduces the same result. This suggests the relational coupling is a scalar-tensor coupling to $T^\mu{}_\mu$ with a geometrically determined coefficient, connecting to standard QFT structure. Moreover, since $\omega_m^2 = a_m/R_m$ and $a_m = c^2/R_m$ is fixed by $n = 4$, the factor ω_m^2 is not a free parameter — it is the same quantity that gives the proton radius. The combination $\kappa\cdot\omega_m^2$ is the only product of closure scales with correct dimensions respecting the modal hierarchy.

[OPEN] Deriving this coupling by constructing matter fields from the same algebraic structure remains the central open problem.

[DERIVED] Linearizing around vacuum:

$$\nabla^2\psi = -\lambda\,\kappa\,\omega_m^2\,\rho_m.$$

Identifying $\psi = -Gm_pM/r$ and using $G = \alpha\kappa\omega_m^2$:

$$\lambda = -4\pi\alpha m_p.$$

[DERIVED] The Newtonian limit reproduces the correct G under the postulated coupling.

7 Cosmological consequences

For homogeneous $\psi(t)$ and flat FLRW metric:

[DERIVED] The Friedmann equations hold with $\rho_\psi = \frac{1}{2}\dot{\psi}^2 + V$ and $p_\psi = \frac{1}{2}\dot{\psi}^2 - V$.

[HYPOTHESIS] Assuming a constant deceleration parameter $q = -\ddot{a}a/\dot{a}^2$ — used here as a phenomenological approximation for the low-redshift regime — one obtains:

$$H(z) = H_0(1+z)^{1+q}, \qquad d_L(z;q) = \frac{(1+z)}{H_0} \cdot \frac{1-(1+z)^{-q}}{q} \quad (q \neq 0).$$

[DERIVED] For $q = -0.55$, deviations from ΛCDM are $\sim 1\%$ at $z \sim 1$, within reach of upcoming surveys (LSST, Euclid, DESI).

8 The open problem of α_G

8.1 Reduction to a single parameter

[DERIVED] With $n = 4$ fixed, the entire gravitational structure reduces to:

$$\alpha_G = \frac{Gm_p^2}{\hbar c} \approx 5.91 \times 10^{-39}.$$

8.2 Reinterpretation as a frequency ratio

[DERIVED] From v4.0:

$$\alpha_G = \left(\frac{\omega_N}{\omega_{\rm Pl}}\right)^2, \qquad \omega_{\rm Pl} = \sqrt{\frac{c^5}{G\hbar}}.$$

The problem reduces to deriving $\omega_{\rm Pl}/\omega_N \approx 1.30 \times 10^{19}$.

8.3 Three routes explored

[OPEN] All three routes fail; each failure is informative.

1. **Individual closure hierarchy.** Maximum suppression from inter-modal ratios ($\omega_V/\omega_N = 1/4$, $\omega_V/\omega_m = 1/\sqrt{15}$) is $\sim 10^{-2}$. Gap to α_G: 37 orders.

2. **Collective closure.** For N nucleons, $\eta(N) = \alpha_G N^{5/3} = 1$ gives $N_{\rm crit} \approx 8.7 \times 10^{22}$ (mass ≈ 0.15 g), with no privileged physical interpretation.

3. **Covariant self-consistency.** The condition $\eta = 1$ at the Planck scale yields an equation with α_G on both sides: circular, does not fix α_G.

8.4 A numerical observation and arithmetic hint

[SPECULATIVE] The appearance of a number of order 10^{39} has a long history in physics, notably in Dirac's large number hypothesis [2]. In the present framework this number emerges as 2^{127}, connecting a deep binary hierarchy to the Planck scale, though the origin of the exponent 127 remains an open question.

Numerically:

$$2^{-127} \approx 5.877 \times 10^{-39}, \qquad \alpha_G \approx 5.906 \times 10^{-39}, \qquad \text{error } 0.5\%.$$

The approximation improves to 0.08% error by including a correction from the $n = 4$ closure:

$$\alpha_G \approx 2^{-127}\left(1 + \frac{1}{240}\right), \qquad 240 = \mathrm{lcm}(16, 15) = \mathrm{lcm}(2^n,\, 2^n - 1)\big|_{n=4}.$$

The numbers 16 and 15 are the squared-frequency denominators of the two modes at $n = 4$. Their least common multiple, 240, is the minimal common period of the two modes — the smallest number of steps after which both return to the same phase simultaneously. The same incommensurability ($15 = 3 \times 5$ being the first composite confinement ratio) was already identified in v4.0 as the origin of proton stability. Both the stability of the proton and the weakness of gravity may thus share a common arithmetic origin.
Caveat: The residual error of 0.08% exceeds the experimental uncertainty of α_G ($\approx$ 0.002%). This is a heuristic hint, not a derivation.

8.5 Precise open question

[OPEN] Does there exist a condition within the generative algebra — perhaps a hierarchy of closures beyond $n = 4$, or a topological constraint — that fixes both $\omega_{\mathrm{Pl}}/\omega_N = 2^{63.5}$ and the correction $1/\mathrm{lcm}(2^n, 2^n - 1)$ exactly, without gravitational input?

9 Other open questions

9.1 Black hole regularity

[SPECULATIVE] Near singularities, the $\beta \neq 0$ regime may regularize the Kretschmann scalar. No exact solution exists; future direction only.

9.2 Cosmic cycles

[SPECULATIVE] Topological changes at extreme scales might lead to cosmic cycles. Highly speculative; not developed here.

10 Falsifiable predictions

1. **[HYPOTHESIS] WEP violations.** The Eötvös parameter should correlate with $\Delta(\kappa\omega_m^2)/(\kappa\omega_m^2)$, not simply with mass difference. MICROSCOPE: $\eta \lesssim 10^{-15}$; STE-QUEST: next target.

2. **[HYPOTHESIS] Scale-dependent G.** Galactic rotation curve anomalies should correlate with $\omega_m^2\rho_m$, not with a fixed acceleration scale a_0 as in MOND [8].

3. **[DERIVED] Low-redshift deviations from ΛCDM.** $d_L(z; q)$ with constant q differs from ΛCDM by 1–3% at $z \sim 1$–2, testable with Type Ia supernovae and gravitational wave standard sirens.

11 Conclusion

We have constructed a covariant extension of the relational theory of gravity. The key advances of this version are:

- **Explicit operator construction:** $(A \times R)^\mu = \epsilon^\mu{}_{\nu\rho\sigma} A^\nu R^\rho n^\sigma$ converts a postulate into a concrete covariant construction.
- **QFT connection:** $\delta S_m/\delta\psi \propto \rho_{\rm rel} \propto T^\mu{}_\mu$ in the non-relativistic limit, with coefficient fixed by the $n = 4$ closure.
- **Corrected tensor contraction:** $P_{\mu\nu}P^{\mu\nu} = -\frac{1}{2}\nabla_\mu\psi\nabla^\mu\psi$ (Appendix A), consistently absorbed into the kinetic coefficient.
- **Systematic analysis of α_G:** Three routes fail. Maximal algebraic suppression is $\sim 10^{-2}$; 37 orders remain unaccounted.
- **Numerical hint:** $\alpha_G \approx 2^{-127}(1 + 1/240)$ with 0.08% error, connecting the weakness of gravity and proton stability through their shared arithmetic incommensurability, in the tradition of Dirac's large number hypothesis.

A notable consequence is that the gravitational scale emerges from the hadronic scale through the binary closure structure, leaving α_G as the only unresolved parameter. The relational framework thus achieves a geometric unification of the proton scale and the gravitational scale, reducing the entire hierarchy of gravity to a single open question: what fixes $\omega_{\rm Pl}/\omega_N = 2^{63.5}$? The answer to this question, once found, would close the circle and reveal the ultimate origin of the weakness of gravity.

A Contraction of $P_{\mu\nu}$

With $P_{\mu\nu} = \frac{1}{2}(n_\mu\nabla_\nu\psi - n_\nu\nabla_\mu\psi)$, $n^\mu n_\mu = -1$, and $n^\mu\nabla_\mu\psi = 0$:

$$\begin{aligned}
P_{\mu\nu}P^{\mu\nu} &= \frac{1}{4}(n_\mu\nabla_\nu\psi - n_\nu\nabla_\mu\psi)(n^\mu\nabla^\nu\psi - n^\nu\nabla^\mu\psi) \\
&= \frac{1}{4}\Big[\underbrace{n_\mu n^\mu}_{-1}(\nabla\psi)^2 - \underbrace{(n^\nu\nabla_\nu\psi)}_{0}(n_\mu\nabla^\mu\psi) - \underbrace{(n^\mu\nabla_\mu\psi)}_{0}(n_\nu\nabla^\nu\psi) + \underbrace{n_\nu n^\nu}_{-1}(\nabla\psi)^2\Big] \\
&= \frac{1}{4}\cdot(-2)\,(\nabla\psi)^2 = -\frac{1}{2}\,\nabla_\mu\psi\nabla^\mu\psi. \qquad (1)
\end{aligned}$$

[DERIVED] The action term $\frac{\beta}{2}P_{\mu\nu}P^{\mu\nu} = -\frac{\beta}{4}(\nabla\psi)^2$ renormalizes the kinetic coefficient: $\frac{1}{2} \to \frac{1}{2} - \frac{\beta}{4} = \frac{1}{2}(1 - \frac{\beta}{2})$. To first order in β, this is equivalent to $\frac{1}{2}(1-\beta)$ used in the main text. For $|\beta| \ll 1$ the correction is small; setting $\beta = 0$ recovers the canonical kinetic term in the weak-field limit.

LLM Disclosure Statement

Models used: Anthropic Claude, OpenAI ChatGPT, Google Gemini, DeepSeek AI.

Employed for: dimensional consistency checking; algebraic stress-testing and detection of the factor-of-2 error corrected in v5.1; systematic exploration of routes for α_G; explicit

Levi-Civita realization; QFT connection of the matter coupling; numerical exploration of the 2^{-127} and $\mathrm{lcm}(16, 15)$ coincidence; language revision.

Estimated contribution: ∼30% drafting/structure; ∼70% human-directed conceptual development and postulate formulation. All mathematical claims were verified by the author independently.

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